

Kaaval Assurance

**Routers predict.
Kaaval verifies.**

An inference assurance plane for open-weight models on AMD compute.

Verify, not predict. Receipts, not promises.

A dark blue background with a glowing green circuit pattern at the bottom. The text "AI INFRASTRUCTURE TOOL" is written in large, white, sans-serif capital letters. Below it is a thin green horizontal line, followed by the text "ACCELERATE. VALIDATE. SCALE." in smaller, white, sans-serif capital letters.

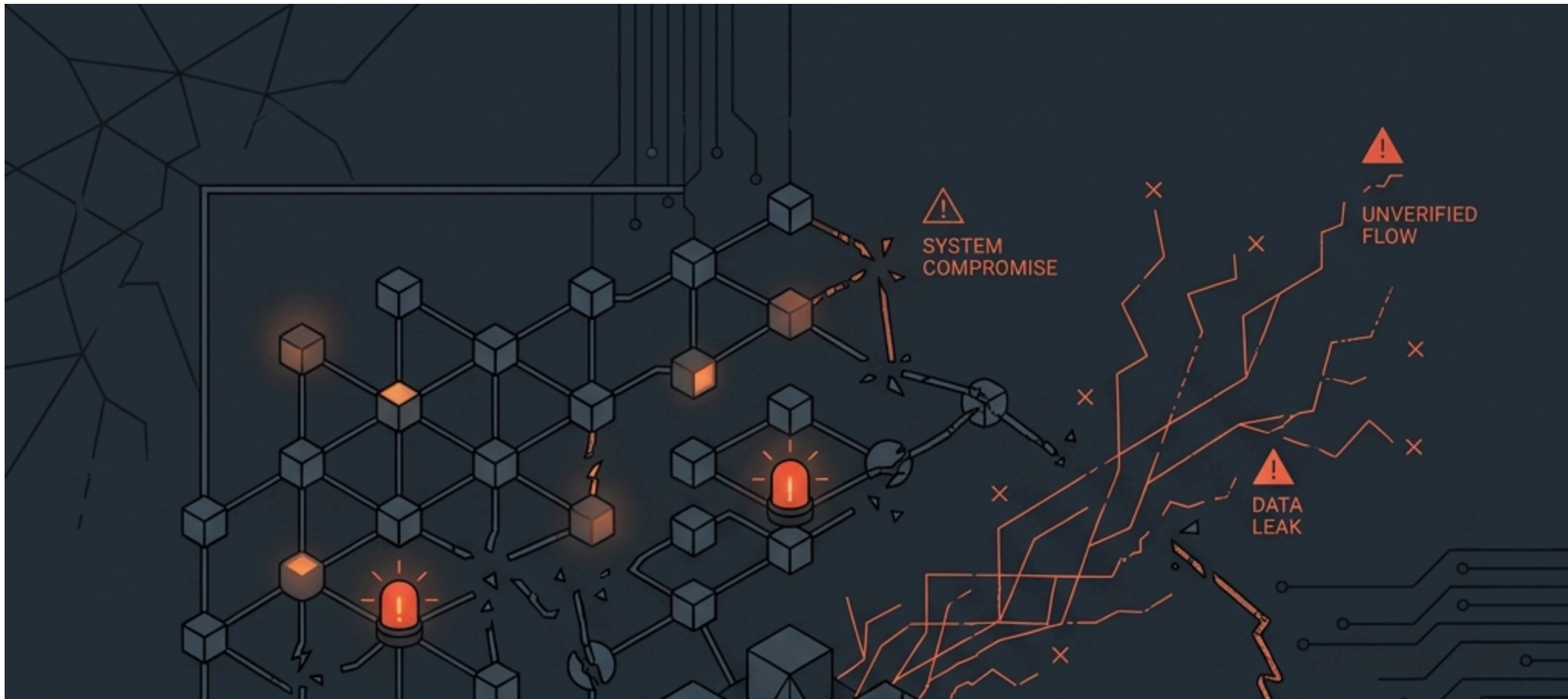
AI INFRASTRUCTURE TOOL

ACCELERATE. VALIDATE. SCALE.

AI Decisions are Liabilities.

Unverified local models approve unauthorized refunds, create exposure, and fail silently.

Standard guardrails check syntax; they cannot verify business logic. As agents handle transactions, quality drift translates directly into financial risk.



Gate. Verify. Escalate.

Local open-weight inference checked by code contracts in <1ms. Only fails safe.

Kaaval intercepts response structures locally. If checks fail, it routes to a remote backup, saving **88%** of API costs while guaranteeing compliance.



3-Layer Assurance Engine

1. Conformance Gate

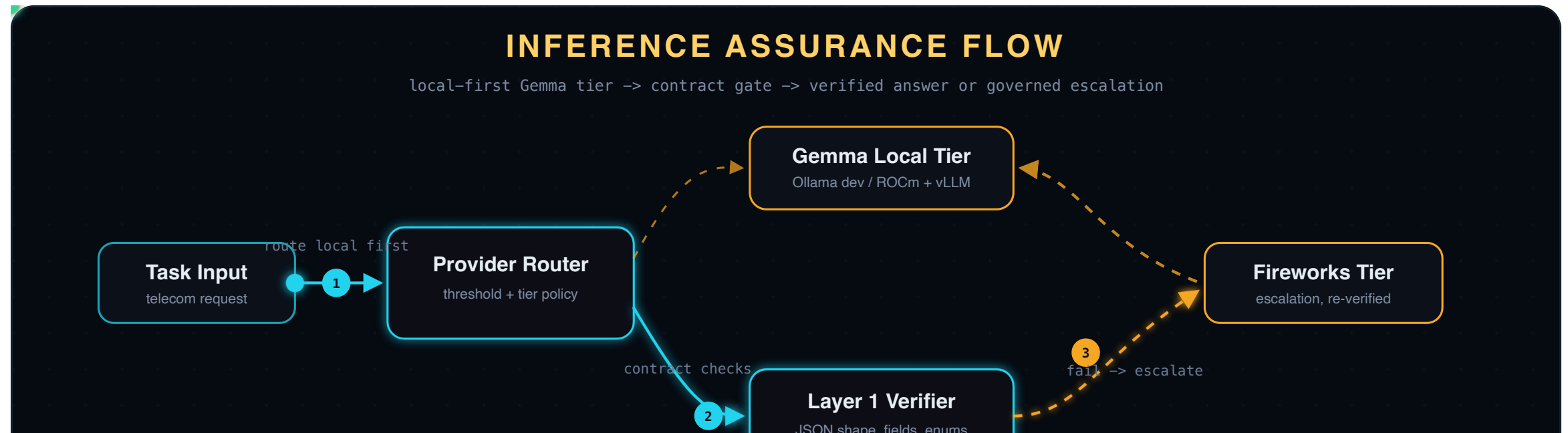
Deterministic JSON shape, enums, & grounding checks in <1ms.

2. Adaptive EWMA Routing

Tracks category failures in real-time. Pre-routes directly to remote on drift.

3. Sampled Offline Audit

Adversarial critic challenges accepted answers with FP-calibration.



Measured Proof on AMD Silicon

100%

[MEASURED]

Contract Conformance

88%

[MEASURED]

API Costs Saved

0%

[MEASURED]

Verify, Don't Predict.

Traditional Routers

Focus on **guessing** model quality *before* generation. Works for content writing, but fails when business contracts cannot tolerate errors.

Kaaval Assurance

Focus on **measuring** contract conformance *after* generation. Built for transactional, policy-bound AI decisions and automated compliance.